

HINT AND CONQUER: REVISITING QUEEN DOMINATION BY SAT SOLVING

1 The Counting Argument

A *line* is a row, column, diagonal, or anti-diagonal of the board. Let D be a dominating set of γ queens. A line is *active* if it contains at least one queen of D , and *empty* otherwise. For $n \geq 3$, the queens on the squares (i, i) with $i \notin \{1, 3\}$ dominate the board, so $\gamma(Q_n) \leq n - 2$ and we assume $\gamma \leq n - 2$ throughout. Since D then occupies at most $n - 2$ columns and at most $n - 2$ rows, it leaves at least two columns and at least two rows empty. Let a and b be the leftmost and rightmost empty columns, and let c and d be the lowest and highest empty rows. The *box* of D is the rectangle of squares (x, y) with $a \leq x \leq b$ and $c \leq y \leq d$, as illustrated in Figure 1. It follows directly from the definition that the four sides of the box (columns a, b and rows c, d) are empty, while every column to the left of a or to the right of b , and every row below c or above d , is active.

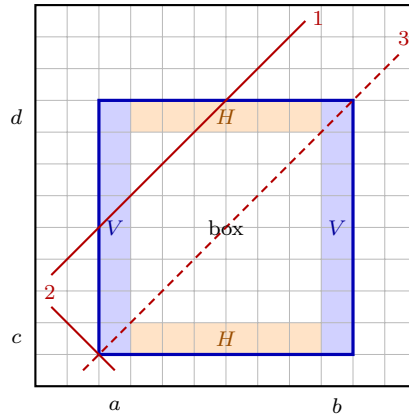


Figure 1: The box of a set of queens (queens not shown). Columns a, b and rows c, d are empty, and every row and column outside the box contains a queen. The vertical sides V (blue) and the horizontal sides H (orange) are shaded while the corners belong to both. Diagonal 1 meets V once, diagonal 2 misses V , and diagonal 3, a corner-to-corner diagonal of a square box, meets V twice.

Let us call the squares of the box in columns a and b its *vertical sides* V , and the squares in rows c and d its *horizontal sides* H . Thus, V contains $2(d - c + 1)$ squares and H contains $2(b - a + 1)$ squares, with the four corners belonging to both.

Every square of V lies in an empty column, so it can only be covered by its row, its diagonal, or its anti-diagonal. In total, V has $2(d - c + 1)$ squares and each needs to be covered.

An active row covers two squares of V if it lies strictly between rows c and d , and none otherwise. The $n - 1 - (d - c)$ rows outside the box are all active, and with γ queens there are at most γ active rows in total. So at most $\gamma - (n - 1 - (d - c))$ active rows lie inside the box.

There are at most γ active diagonals and at most γ active anti-diagonals. Each such line contains at most one square in column a and at most one in column b , and hence meets zero,

one, or two squares of V . It is useful to think of one square per active diagonal or anti-diagonal as the baseline, corresponding to coverage without waste. Such a line that misses V wastes one unit, while one that meets V twice provides one extra unit that can compensate for such waste.

At the baseline, the rows can cover at most $2(\gamma - n + 1 + d - c)$ squares of V , while the active diagonals and anti-diagonals together contribute at most 2γ baseline units. Since V contains $2(d - c + 1)$ squares, this leaves only $2(\gamma - n + 1 + d - c) + 2\gamma - 2(d - c + 1) = 2(2\gamma - n) = 2\delta$ units available for waste. Each diagonal or anti-diagonal that meets V twice provides one additional unit. Thus, the total waste is limited by a small *budget* 2δ plus the number of such lines. The following lemma quantifies exactly how much waste can be afforded.

Lemma 1. *Let D be a dominating set of at most γ queens on the $n \times n$ board that leaves at least two rows and two columns empty. For the vertical sides V , let*

$$\text{over}(V) = \sum_{z \in V} (\text{number of active lines through } z - 1),$$

and let $\text{miss}(V)$ and $\text{twice}(V)$ denote, respectively, the number of active diagonals and anti-diagonals that meet V zero and two times. Define

$$e_{\text{row}} = \gamma - (\text{number of active rows}),$$

and define e_{col} , e_{diag} , and e_{anti} analogously. These excesses are non-negative, since at most γ queens can activate at most γ lines of each type. Define $\text{over}(H)$, $\text{miss}(H)$, and $\text{twice}(H)$ analogously for the horizontal sides H . Then

$$\text{over}(V) + \text{miss}(V) + 2e_{\text{row}} + e_{\text{diag}} + e_{\text{anti}} = \text{twice}(V) + 2\delta, \quad (1)$$

$$\text{over}(H) + \text{miss}(H) + 2e_{\text{col}} + e_{\text{diag}} + e_{\text{anti}} = \text{twice}(H) + 2\delta. \quad (2)$$

Proof. We count the active lines through the squares of V in two ways.

By squares. For each square of V , count the number of active lines through it. Since D is dominating and the two columns containing V are empty, every square has at least one such line. Counting one active line for each of the $2(d - c + 1)$ squares gives $2(d - c + 1)$, while all additional active lines are counted by $\text{over}(V)$. Thus, the total is $2(d - c + 1) + \text{over}(V)$.

By lines. The columns containing V are empty and therefore contribute nothing. There are $\gamma - e_{\text{row}}$ active rows in total. Of these, exactly $n - 1 - (d - c)$ lie outside the box, leaving $\gamma - e_{\text{row}} - (n - 1 - (d - c))$ active rows strictly inside the box. Each of these meets V twice, so the rows contribute $2(\gamma - e_{\text{row}} - n + 1 + d - c)$.

There are $\gamma - e_{\text{diag}}$ active diagonals and $\gamma - e_{\text{anti}}$ active anti-diagonals. Taking one square of V per active line as the baseline, a line that misses V contributes one fewer, while a line that meets V twice contributes one additional square. Thus, the diagonals and anti-diagonals contribute $(\gamma - e_{\text{diag}}) + (\gamma - e_{\text{anti}}) - \text{miss}(V) + \text{twice}(V)$.

Equating the two counts gives $2(d - c + 1) + \text{over}(V) = 2(\gamma - e_{\text{row}} - n + 1 + d - c) + (\gamma - e_{\text{diag}}) + (\gamma - e_{\text{anti}}) - \text{miss}(V) + \text{twice}(V)$. Rearranging and using $\delta = 2\gamma - n$ gives $\text{over}(V) + \text{miss}(V) + 2e_{\text{row}} + e_{\text{diag}} + e_{\text{anti}} = \text{twice}(V) + 2\delta$, which is (1). Equation (2) follows by the same argument with rows and columns exchanged. $\square$

For a fixed box, let T_V be the set of diagonals and anti-diagonals that contain two squares of V , and define T_H analogously for H . Unlike $\text{twice}(V)$ and $\text{twice}(H)$, these sets depend only on the box. Let L_ℓ indicate whether line ℓ is active. Then $\text{twice}(V)$ and $\text{twice}(H)$ count the active lines in T_V and T_H , respectively. Thus, $\text{twice}(V) = \sum_{\ell \in T_V} L_\ell$ and $\text{twice}(H) = \sum_{\ell \in T_H} L_\ell$. Consequently,

$$|T_V| - \text{twice}(V) = \sum_{\ell \in T_V} (1 - L_\ell) \quad \text{and} \quad |T_H| - \text{twice}(H) = \sum_{\ell \in T_H} (1 - L_\ell),$$

which count the lines that could meet the corresponding sides twice but are inactive. Moving $\text{twice}(V)$ and $\text{twice}(H)$ to the left in Equations (1) and (2), and adding $|T_V|$ and $|T_H|$, respectively, gives

$$\text{over}(V) + \text{miss}(V) + 2e_{\text{row}} + e_{\text{diag}} + e_{\text{anti}} + \sum_{\ell \in T_V} (1 - L_\ell) = |T_V| + 2\delta, \quad (3)$$

$$\text{over}(H) + \text{miss}(H) + 2e_{\text{col}} + e_{\text{diag}} + e_{\text{anti}} + \sum_{\ell \in T_H} (1 - L_\ell) = |T_H| + 2\delta. \quad (4)$$

In this form, the left sides consist entirely of waste, while the budgets on the right depend only on the box and δ . The main purpose of this rewrite is that the resulting constraints are easier to encode in SAT than their original forms.

By the geometry of the box, $T_V = \emptyset$ if the box is wider than tall, $T_H = \emptyset$ if it is taller than wide, and $|T_V| = |T_H| = 2$ if it is square. In particular, when $\gamma = n/2$, we have $\delta = 0$, so the budgets reduce to $|T_V|$ and $|T_H|$. If the box is wider than tall, the budget for V is therefore zero, if it is taller than wide, the budget for H is zero, and when the box is square, both budgets are two. Thus, only a very small amount of waste is possible. In this sense, a dominating set of $n/2$ queens is virtually a set of non-attacking queens whose lines cover the sides of the box exactly once. Our main conjecture is that this rigidity is difficult for a SAT solver to discover on its own, and that efficiently encoding this counting argument may provide useful hints to the solver. Thus, the rest of the work seeks an efficient encoding of this argument and investigates its impact on solver performance empirically.

2 SAT-Based Methodology

For an $n \times n$ board and a number of queens γ , we generate one formula for every possible box, which decides whether γ queens can dominate the board with that fixed box. Every dominating set has exactly one box, so the formulas for different boxes describe disjoint sets of solutions and together cover all possible solutions. Some boxes can be eliminated before generating the formulas. In particular, boxes with more than γ columns outside or more than γ rows outside are impossible, since each such column or row requires its own queen. Moreover, reflecting or transposing the board maps symmetric boxes to each other, so we keep only one box from each symmetry class. Each remaining formula consists of a core encoding of domination in terms of squares and lines, additional clauses immediately implied by the fixed box, and the counting hints from Section 1.

2.1 Domination encoding

For every square i , let Q_i indicate whether a queen is placed on i , and for every line ℓ , let L_ℓ indicate whether ℓ is active. Let $L(i)$ denote the set consisting of the row, column, diagonal, and anti-diagonal passing through square i . Our encoding is based on the line-variable encoding of Rostami and Bright. For every line ℓ , we add

$$\neg L_\ell \vee \bigvee_{i \in \ell} Q_i,$$

so that an active line contains a queen. For our counting constraints, we also need the converse relation. Thus, for every square i and every $\ell \in L(i)$, we add

$$\neg Q_i \vee L_\ell.$$

Consequently, L_ℓ is true exactly when ℓ contains a queen. Since a square is dominated whenever at least one of its four lines contains a queen, for every square i we add

$$\bigvee_{\ell \in L(i)} L_\ell.$$

To bound the size of the dominating set, we impose

$$\sum_i Q_i \leq \gamma.$$

Moreover, since each queen belongs to exactly one row, one column, one diagonal, and one anti-diagonal, at most γ lines of each type can be active. Thus, writing $\mathcal{R}$, $\mathcal{C}$, $\mathcal{D}$, and $\mathcal{A}$ for the sets of rows, columns, diagonals, and anti-diagonals, respectively, we impose

$$\sum_{\ell \in \mathcal{R}} L_\ell \leq \gamma, \quad \sum_{\ell \in \mathcal{C}} L_\ell \leq \gamma, \quad \sum_{\ell \in \mathcal{D}} L_\ell \leq \gamma, \quad \sum_{\ell \in \mathcal{A}} L_\ell \leq \gamma.$$

2.2 Box encoding

For a fixed box with empty columns $a < b$ and empty rows $c < d$, we first force its four sides to be empty. Thus, for every square i in column a , column b , row c , or row d , we add

$$\neg Q_i.$$

Every column to the left of a or to the right of b , and every row below c or above d , must be active. For each such line ℓ , we add

$$\bigvee_{i \in \ell} Q_i \quad \text{and} \quad L_\ell.$$

2.3 Counting hints

We now encode the counting constraints (3) and (4). Since the box is fixed, we can determine in advance which diagonals and anti-diagonals miss each pair of sides and which meet them twice. For each side, every term on the left-hand side is represented by literals that become true when the corresponding waste occurs. We then constrain the number of these literals by the budget on the right-hand side.

Over-coverage. For every square z of V , let $L_V(z)$ be the set of lines through z other than the empty column containing it. Thus, $|L_V(z)| = 3$, except at the corners of the box, where the empty row is also excluded and $|L_V(z)| = 2$. We introduce a variable $o_{z,2}$ that is forced to be true whenever at least two lines in $L_V(z)$ are active. For every pair $\ell, \ell' \in L_V(z)$, we therefore add

$$L_\ell \wedge L_{\ell'} \rightarrow o_{z,2}.$$

If $|L_V(z)| = 3$, say $L_V(z) = \{\ell_1, \ell_2, \ell_3\}$, we additionally introduce $o_{z,3}$ and add

$$L_{\ell_1} \wedge L_{\ell_2} \wedge L_{\ell_3} \rightarrow o_{z,3}.$$

Thus, a square covered by two active lines contributes one over-coverage literal, while a square covered by three active lines contributes two. These literals together count $\text{over}(V)$. The corresponding literals for H are defined analogously.

Missing lines. For a fixed box, we know which diagonals and anti-diagonals do not meet V . For each such line ℓ , the variable L_ℓ itself serves as a waste literal, since it contributes to $\text{miss}(V)$ exactly when the line is active. The same construction is used for H .

Excess. The row excess is at least j exactly when at least $n - \gamma + j$ rows are inactive. We therefore use a literal for each such threshold, so that the number of true threshold literals represents the row excess. The column, diagonal, and anti-diagonal excesses are encoded analogously.

Final constraints. Recall that T_V is the set of diagonals and anti-diagonals that can meet V twice. For each $\ell \in T_V$, the literal $\neg L_\ell$ contributes one unit of waste in (3). We therefore collect the over-coverage literals for V , the active-line literals for diagonals and anti-diagonals that miss V , the row-excess literals twice, the diagonal- and anti-diagonal-excess literals once, and $\neg L_\ell$ for every $\ell \in T_V$. We impose

$$\sum_{w \in W_V} w \leq |T_V| + 2\delta,$$

where W_V denotes this multiset of waste literals. Listing the row-excess literals twice gives them weight two without requiring a weighted pseudo-Boolean constraint. The constraint for H is constructed analogously, with column excess in place of row excess:

$$\sum_{w \in W_H} w \leq |T_H| + 2\delta.$$

3 Results

For $n = 26$ and $n = 28$, we obtain the new values $\gamma(Q_{26}) = 14$ and $\gamma(Q_{28}) = 15$. Both results were obtained in under a few minutes.